Supporting Information

Molecular Simulation of Doravirine Resistance in HIV-1 Reverse Transcriptase

Rohit Satija, Kaiming Tao, Robert W. Shafer
Division of Infectious Diseases, School of Medicine, Stanford University, Stanford, CA, USA
Corresponding author: rsatija@stanford.edu

Contents

1 Is the RT-DOR crystal structure representative of the true RT-DOR structural ensemble?

2 Sensitivity of the MM/GBSA energies to the ligand charge model

3 The crystallographic pose is recovered from perturbed starting poses

4 Convergence of the molecular dynamics ensemble

5 Sensitivity of the MM/GBSA energies to the snapshot relaxation protocol

6 Non-equilibrium switching work distributions

7 Finite-size corrections for the charge-changing alchemical legs

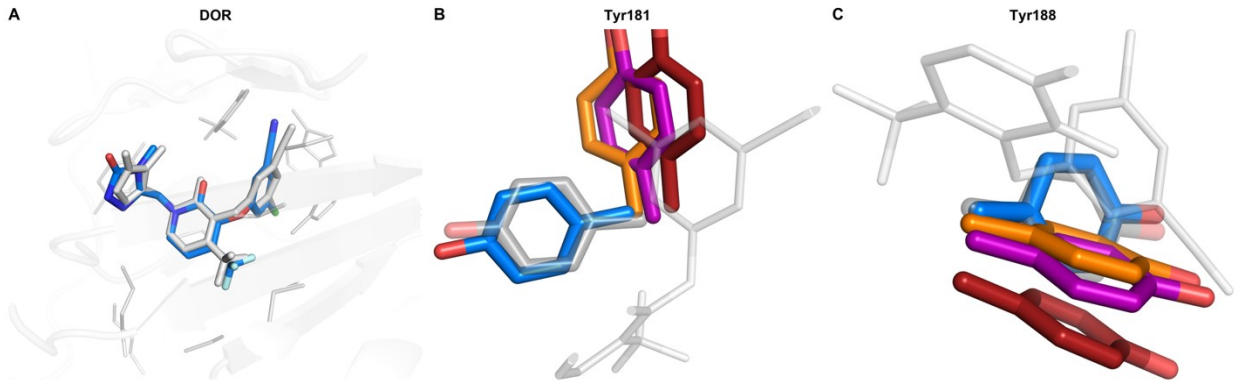
8 Protein-level conformational analysis

9 Conformation of the unliganded binding pocket

1. Is the RT-DOR crystal structure representative of the true RT-DOR structural ensemble?

We address this question in two steps: first by comparing the DOR pose in the crystal structure (PDB ID: 4NCG) with a second, independent RT-DOR-DNA ternary structure determined by cryo-EM (PDB ID: 7Z2G), and then by quantifying changes in key residues of the NNIBP, using other published structures of NNRTI complexes as a reference set. For all comparisons, we align 16 NNIBP-lining Cα atoms of 4NCG, fifteen in p66 and Glu138 in p51. 7Z2G is a 3.65 Å single-particle cryo-EM structure of HIV-1 RT bound to a 38-mer DNA aptamer and DOR, determined by a different group, by a different method, and in a different functional state from the 2.58 Å X-ray structure 4NCG. Superposed on the pocket Cα atoms alone, DOR agrees between them to 0.62 Å heavy-atom RMSD over its 29 atoms, with a centroid displacement of 0.30 Å (Supplementary Figure 1A). Fitted ligand on ligand, its internal conformation agrees to 0.44 Å, no ring system is placed differently from the rest, and the four rotatable dihedrals agree to within 15°. The pocket agrees to 0.49 Å over the same Cα atoms and to a median side-chain RMSD of 0.80 Å, and the NNIBP volume is 198 Å³ in 4NCG against 194 Å³ in 7Z2G. Per-residue values are in Supplementary Table 7. Furthermore, even without any superposition, the minimum heavy-atom distance from each NNIBP residue to DOR agrees between the two structures to a median of 0.16 Å, with the largest discrepancy of 0.41 Å at Lys103 and a correlation of r = 0.974 across the sixteen residues.

To calibrate these alignment results, we applied the identical superposition to three further complexes of wild-type RT with an NNRTI: RPV (2ZD1, 1.80 Å), EFV (1FK9, 2.50 Å) and NVP (1VRT, 2.20 Å). Tyr181 occupies a different rotamer in the DOR structures than in all three others: its χ1 angle differs from 4NCG by 93°, 95° and 90° respectively, displacing the ring centroid by 4.9, 4.2 and 4.2 Å (Supplementary Figure 1B). The NNIBP volume across the five structures ranges from 181 to 258 Å³. Furthermore, Tyr188 χ1 differs by only 13–20° and its ring centroid by 1.3–2.6 Å across all five structures (Supplementary Figure 1C). Per-structure values for both tyrosines are in Supplementary Table 8. Taken together these results support the claim that the DOR-containing NNIBP conformations are the same in two experimental structures determined independently, while the same pocket demonstrably adopts a different conformation when it holds other drugs in the same class. The starting geometry of this work is therefore a property of the DOR–RT complex rather than an artifact of one crystal form.



Supplementary Figure 1. The DOR-NNIBP pose is reproducible across independently determined RT-DOR structures. **(A)** Doravirine in the two independent experimental structures, 4NCG (X-ray, 2.58 Å, grey) and 7Z2G (cryo-EM, 3.65 Å, blue), with the 4NCG pocket side chains in faint grey; heavy-atom RMSD 0.62 Å over 29 atoms. **(B)** Tyr181 in five NNRTI complexes: 4NCG (grey) and 7Z2G (blue) against rilpivirine 2ZD1 (red), efavirenz 1FK9 (orange) and nevirapine 1VRT (purple). The two DOR structures agree with each other and differ from the other three by 90–95° in χ1. **(C)** Tyr188 in the same five structures, which share one rotamer and differ by only 13–20°. Doravirine from 4NCG is drawn faintly in (B) and (C) for orientation.

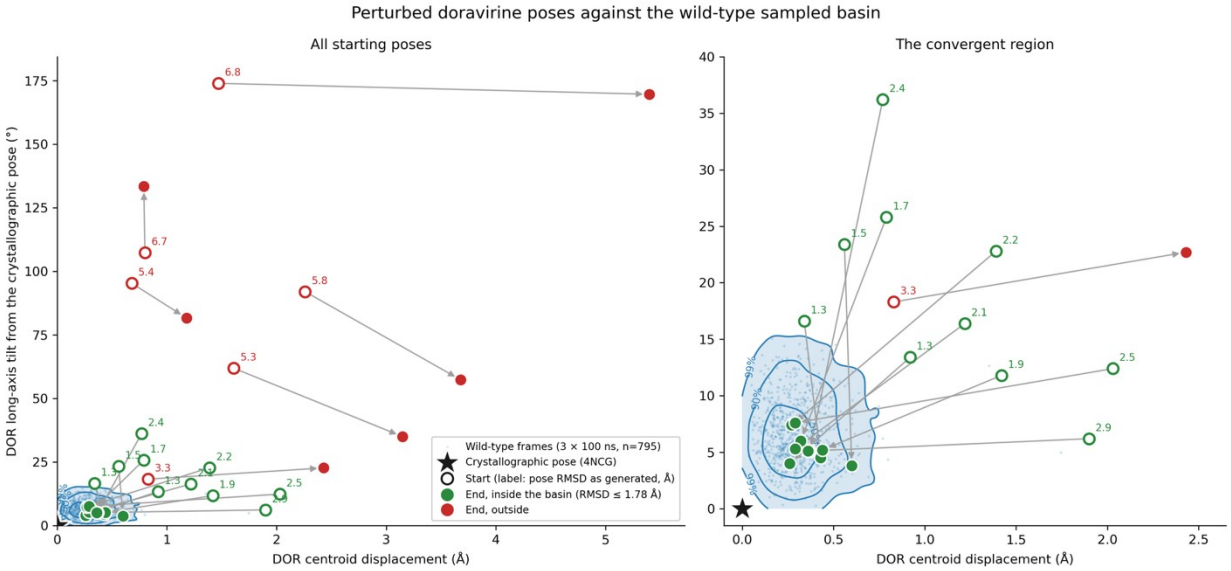
2. Sensitivity of the MM/GBSA energies to the ligand charge model

DOR parameters for the equilibrium MD simulations were generated with Gasteiger partial charges, as opposed to AM1-BCC for the FEP simulations. To test how much that choice matters, the same trajectories and snapshots were re-scored with AM1-BCC charges for wild type and four genotypes, chosen to span the panel's sensitivity to the charge assignment: G190E and K103N+M230L, where the electrostatic and generalised-Born terms contribute most to ΔΔETotal, together with V106A and Y188L, where they contribute least. The two charge sets are both neutral and correlate at 0.890, but AM1-BCC is more polarised (mean |q| 0.228 e against 0.138 e; RMS difference 0.165 e, maximum 0.390 e), with the largest differences on the triazolinone and pyridinone carbonyl and amide atoms. The WT-referenced ΔΔETotal changed by at most 1.33 kcal/mol and kept its sign for every genotype tested (Supplementary Table 5). The individual components move further than the total and in opposing directions: for K103N+M230L the electrostatic term shifts by −2.13 kcal/mol and the generalised-Born term by +2.01 kcal/mol, which largely cancel.

3. The crystallographic pose is recovered from perturbed starting poses

To test whether representative systems from independent starting poses can recover the same DOR-bound state used in the simulations, we ran energy minimization followed by 2 ns MD equilibrations for a ladder of sixteen starting poses at increasing distance from the

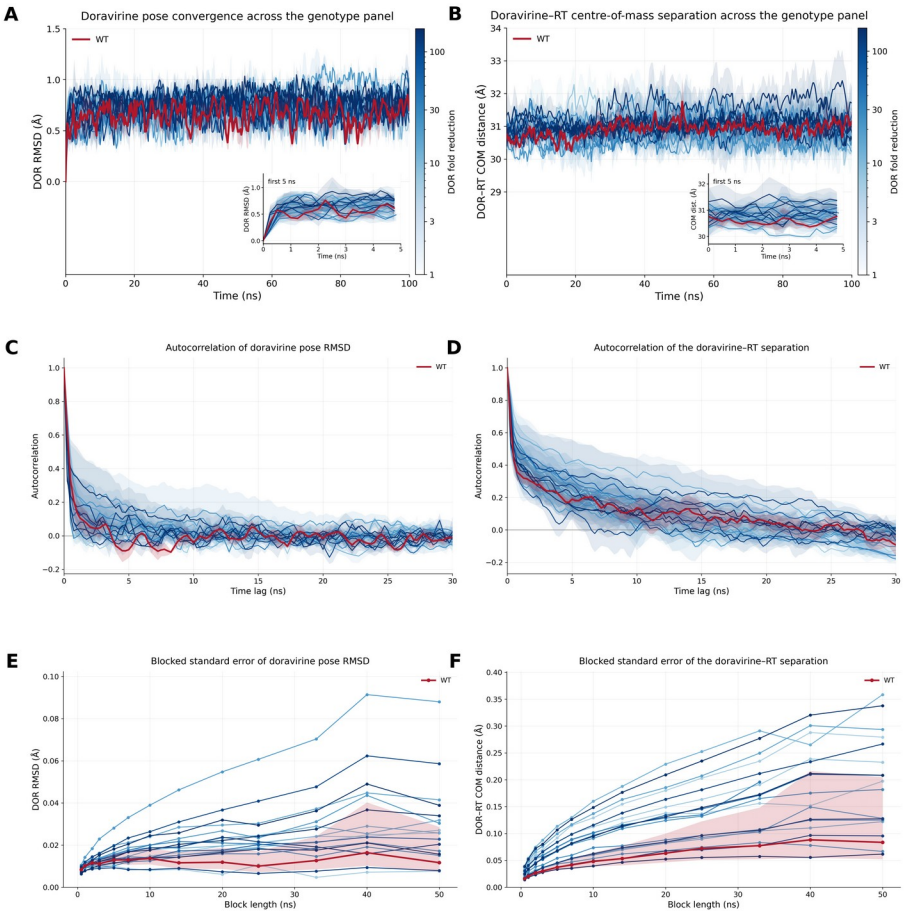
crystallographic pose. In the wild-type (WT) production ensemble, the DOR pose RMSD lies between 0.33 and 1.78 Å, the centroid displacement is at most 1.75 Å and the long-axis tilt at most 15.1°; we take that envelope as the sampled WT basin. All ten poses starting between 1.3 and 2.9 Å converged into the WT basin, ending at a median pose RMSD of 0.74 to 1.01 Å with 78 to 100% of their frames inside the envelope. However, extreme perturbations, corresponding to the six poses starting at 3.3 Å or beyond, failed to converge to the WT basin in the 2 ns simulations, ending between 5.0 and 9.4 Å (Supplementary Figure 2). The crystallographic pose is therefore an attractor with a basin of, at least, roughly 3 Å DOR pose RMSD from the crystal structure used in this study.



Supplementary Figure 2: Perturbed doravirine poses against the wild-type sampled basin. Axes are the pocket-frame coordinates after superposition on the NNIBP Cα atoms: centroid displacement separates translation, long-axis tilt separates rotation. Blue shading and contours give the density of the wild-type production ensemble (3 × 100 ns) enclosing 50, 90 and 99% of frames; the star is the crystallographic pose. Each run is drawn as an arrow from its starting pose, labelled with its heavy-atom RMSD as generated, to the median of its second half (last 1ns of the equilibration simulation). Green ends inside the basin, red outside. (A) All sixteen poses. (B) The same data over the convergent region.

4. Convergence of the molecular dynamics ensemble

Three 100 ns replicates are long relative to local relaxation of the DOR pose but short relative to RT domain rearrangements and the conformational equilibrium of the unliganded pocket. Production replicates were seeded from the same crystallographic pose with small coordinate perturbations. Stable RMSD values, overlapping work distributions and small replicate-to-replicate errors therefore establish convergence within the sampled basin, not that it is the only relevant basin. Across the panel, the DOR heavy-atom RMSD from the crystallographic pose settles between 0.64 and 0.93 Å, with WT at 0.66 Å. The separate perturbed-pose tests are reported in Supplementary Section 3 and Supplementary Figure 2.

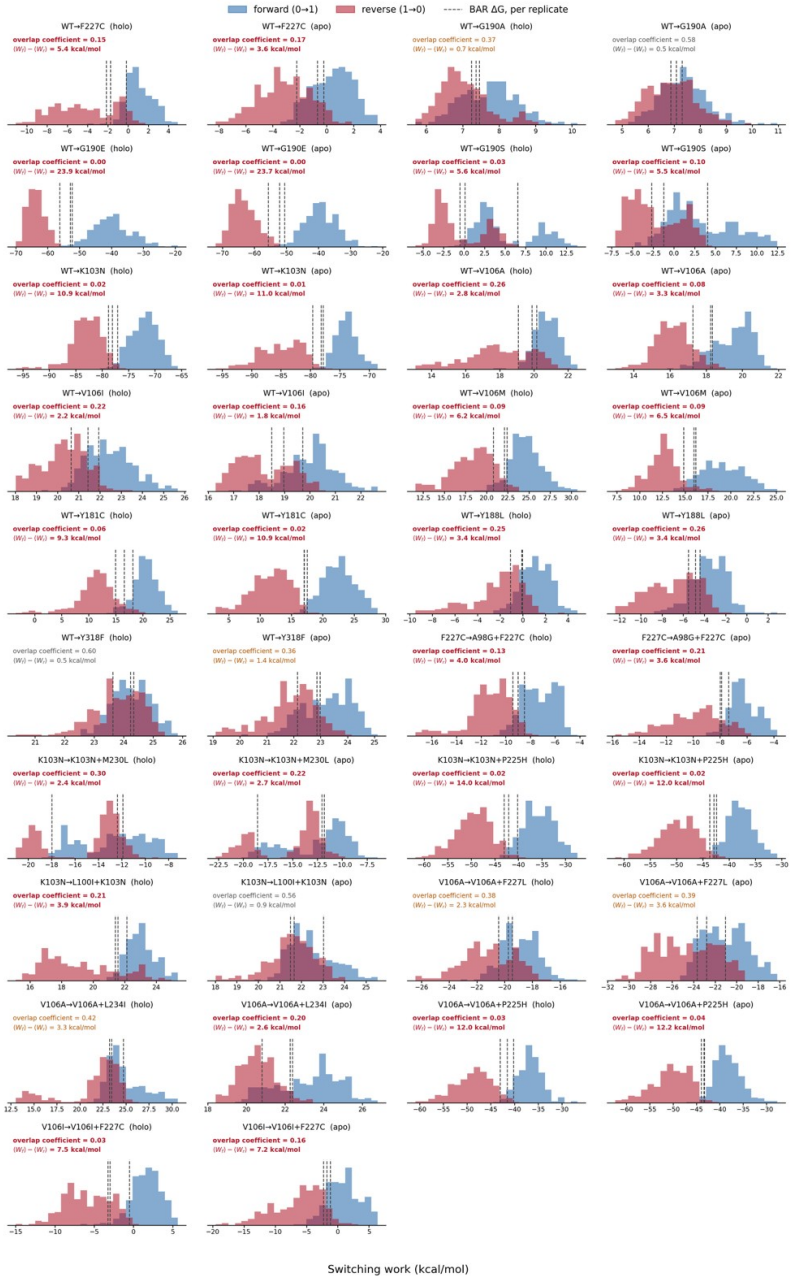


Supplementary Figure 3. DOR Root Mean Squared Deviation (RMSD) and distance between centers of mass (COM) of DOR and RT indicate 100ns runtime is sufficient for DOR-RT interactions to stabilize. (A) Each solid line provides the average RMSD of the DOR heavy atoms vs the minimized structure within the NNIBP (at t = 0) computed at each time point across triplicate simulations. Shaded regions represent the standard error of this mean (SEM). The red curve represents WT and the remaining trajectories are color-coded based on the respective genotype's fold-change in susceptibility (Table 1 in main text). Inset shows the same data highlighting the rapid changes in the first 5ns of the simulations. **(B)** Similar plot for the average distance between COMs of DOR and RT atoms. **(C, D)** Normalised autocorrelation functions of these two chosen convergence metrics, computed for each replicate and averaged within each genotype, with the shaded region showing the SEM across replicates. From these plots, DOR RMSD and the DOR-RT COM distance in our simulations decorrelate within about 5 ns and roughly 20 ns, respectively. **(E, F)** Blocked standard error of the convergence metrics against block length. The shaded band, drawn for wild type alone to keep the panel legible, is the 95% chi-square interval on a standard error estimated from that number of blocks. From these plots, the blocked error is nearly flat beyond 10 ns for DOR RMSD, but DOR-RT COM distance takes about 40-50 ns, up to the statistical limit of the trajectories simulated here.

5. Sensitivity of the MM/GBSA energies to the snapshot relaxation protocol

Each snapshot is energy-minimized before its energy is evaluated. We compared hydrogen-only relaxation, with heavy atoms restrained, against the unrestrained full minimization used in the main text, for all 19 mutant genotypes (Supplementary Table 6). The two protocols agree on the terms that do not depend on close atomic contacts: the Spearman correlation between them is 0.69 for $\Delta\Delta E_{elec}$ and 0.81 for $\Delta\Delta E_{GB}$. This is not the case for $\Delta\Delta E_{vdW}$ (0.26) or $\Delta\Delta E_{Total}$ (0.09). We find that hydrogen-only relaxation leaves heavy-atom contacts in place, and those contacts enter the van der Waals term, which dominates total MM/GBSA computed interaction energy shifts. Furthermore, over the 13 genotypes with an established phenotype, $\Delta\Delta E_{Total}$ from full minimization correlates with $\log_{10}$ fold reduction in DOR susceptibility at Spearman +0.45 ($p = 0.13$), while hydrogen-only relaxation gives -0.32 ($p = 0.29$), though neither reaches statistical significance. For completeness, we also tested single-point evaluation, with no relaxation at all, on six genotypes. In these cases, we found that $\Delta\Delta E_{Total}$ is dominated by a small number of snapshots carrying steric contacts, reaching 7×10^8 kcal/mol in the worst case. Our view is $\Delta\Delta E_{Total}$ is not interpretable until close heavy atom contacts are relieved.

6. Non-equilibrium switching work distributions

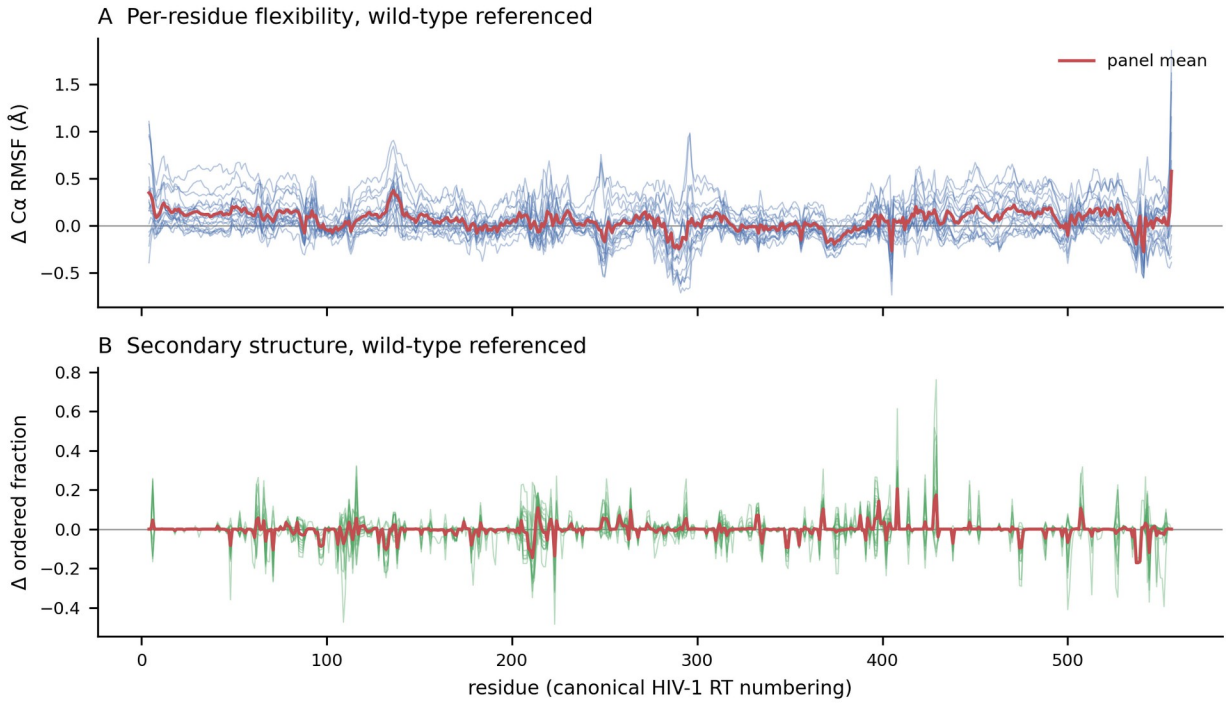


Supplementary Figure 4. Non-equilibrium switching work distributions for all alchemical transformations. Forward ($0 \rightarrow 1$, blue) and reverse ($1 \rightarrow 0$, red) switching work for each of the 19 alchemical legs, shown separately for the *holo* and *apo* phases and pooled across three independent replicates (100 switches per direction per replicate). Dashed lines mark the BAR free energy, ΔG, of each replicate. Each panel reports the forward/reverse overlap coefficient — the shared area under the two normalized work densities, where 0 is disjoint and 1 identical — and the hysteresis $\langle W_f \rangle - \langle W_r \rangle$, which measures the work dissipated by switching at finite speed. Annotations are colored red below an overlap of 0.3 and amber below 0.5. Compound genotypes are reached by two sequential legs, both shown.

7. Finite-size corrections for the charge-changing alchemical legs

The leading periodicity (net-charge self-energy) term of the Rocklin scheme is +0.0407 kcal/mol per phase and cancels to below 0.0001 kcal/mol in $\Delta\Delta G_{bind}$ for both WT → K103N and WT → G190E. The higher-order terms — net-charge undersolvation, discrete-solvent and residual integrated potential — were not applied, since computing them exactly requires a Poisson-Boltzmann treatment of each environment. However, the charge change is identical in the holo and apo phases by construction; and the cells are matched to within 0.13% (K103N) and 0.31% (G190E) in volume, with DOR occupying 0.019% of the cell. Taking a per-phase correction of 1 kcal/mol, about 25 times the leading term, the residual on ΔΔGbind is below 0.004 kcal/mol, at least 40-fold smaller than the 0.16–1.06 kcal/mol per-leg standard errors derived from Supplementary Table 3.

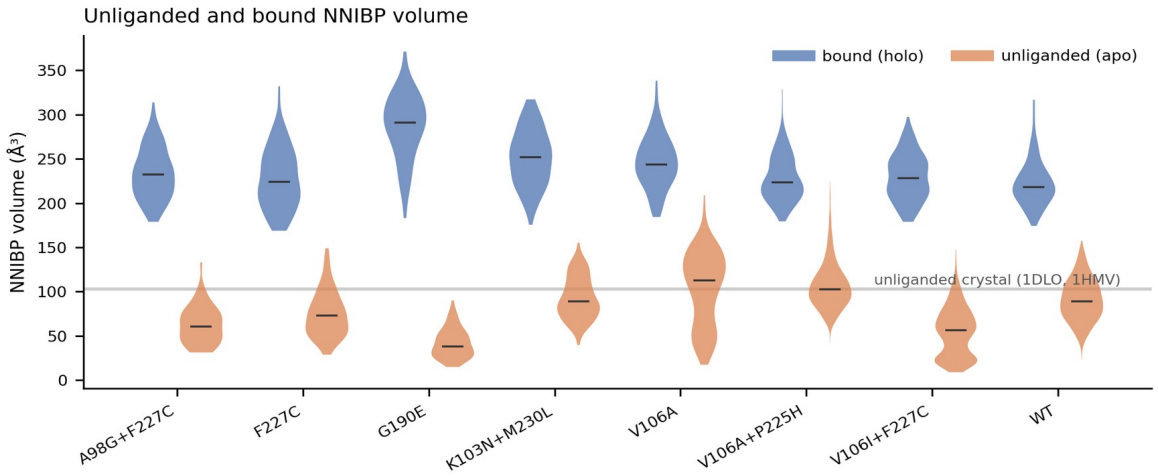
8. Protein-level conformational analysis



Supplementary Figure 5. Protein-level conformational analysis. Per-residue Cα RMSF after superposition on the NNIBP-lining residues, and three-state secondary structure by DSSP, both referenced to wild type across the panel. Of the 171 genotype-by-region fluctuation comparisons, 22 have a 95% confidence interval excluding zero, and they are not scattered at random. The primer grip (225–235) becomes more mobile in eleven of the nineteen genotypes, by 0.07 to 0.17 Å and always in the same direction, and G190E is more mobile than wild type in five regions at once: the pocket lining, the primer grip, β6, the p66 palm, and the p66 fingers, the last by 0.56 Å [0.12, 0.99]. Secondary structure moves by at most 0.11 of strand character, in the YMDD motif of K103N, and only 9 of 342 such comparisons are resolved at three replicates. No genotype remodels the fold in the bound state, so the differences in Table 3 are local to the DOR interface rather than consequences of broader conformational change. The one systematic signal outside the pocket is the loosening of the primer grip, which a binary complex containing no template/primer cannot connect to a mechanism.

9. Conformation of the unliganded binding pocket

Three 100 ns apo replicates were run for WT and all nineteen simulated mutant genotypes, including the F227C alchemical intermediate. Supplementary Table 4 reports pocket volumes in both states, the per-replicate values and the WT-referenced bound–unliganded differences alongside the other structural observables. Supplementary Figure 6 shows the volume distributions. These measurements describe pocket geometry; they do not quantify the free energy required to form the pocket.



Supplementary Figure 6. Apo and holo NNIBP volume distributions. Per-frame pocket volume for the bound state over full production, and for the unliganded state over a matched 100 ns window. The unliganded pocket is much smaller than the bound pocket in every genotype; per-genotype values are given in Supplementary Table 4. In one G190E apo replicate RT became extended, from 21.5 ns onward, with the chain spanning more than the box edge while every covalent bond stayed intact at 1.2 Å. Those 158 of 200 frames are excluded; the other two G190E replicates and every other apo replicate in the panel keep an intact pocket throughout, and no holo trajectory behaves this way.